

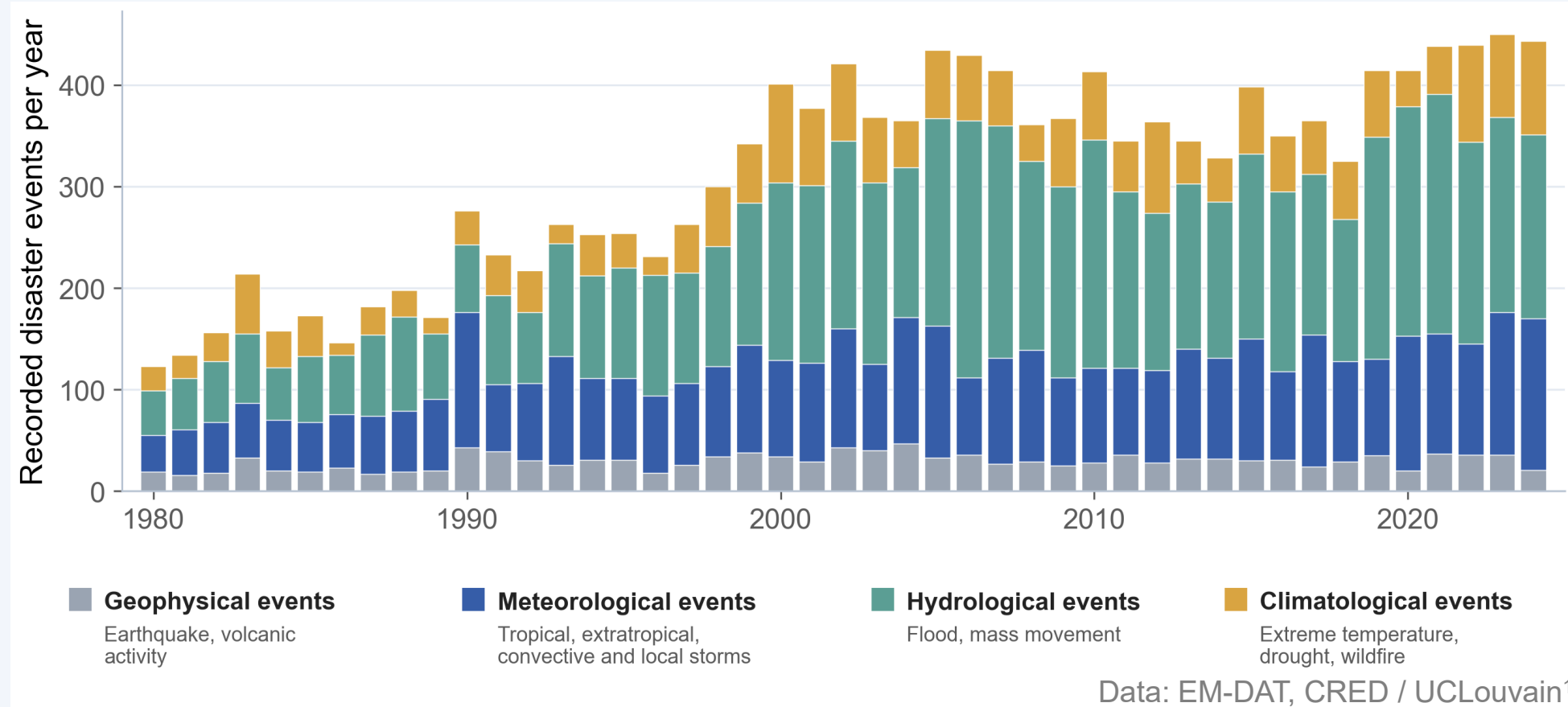
Assessing the Climate Resilience of Electricity Networks on DAFNI



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Background and Context

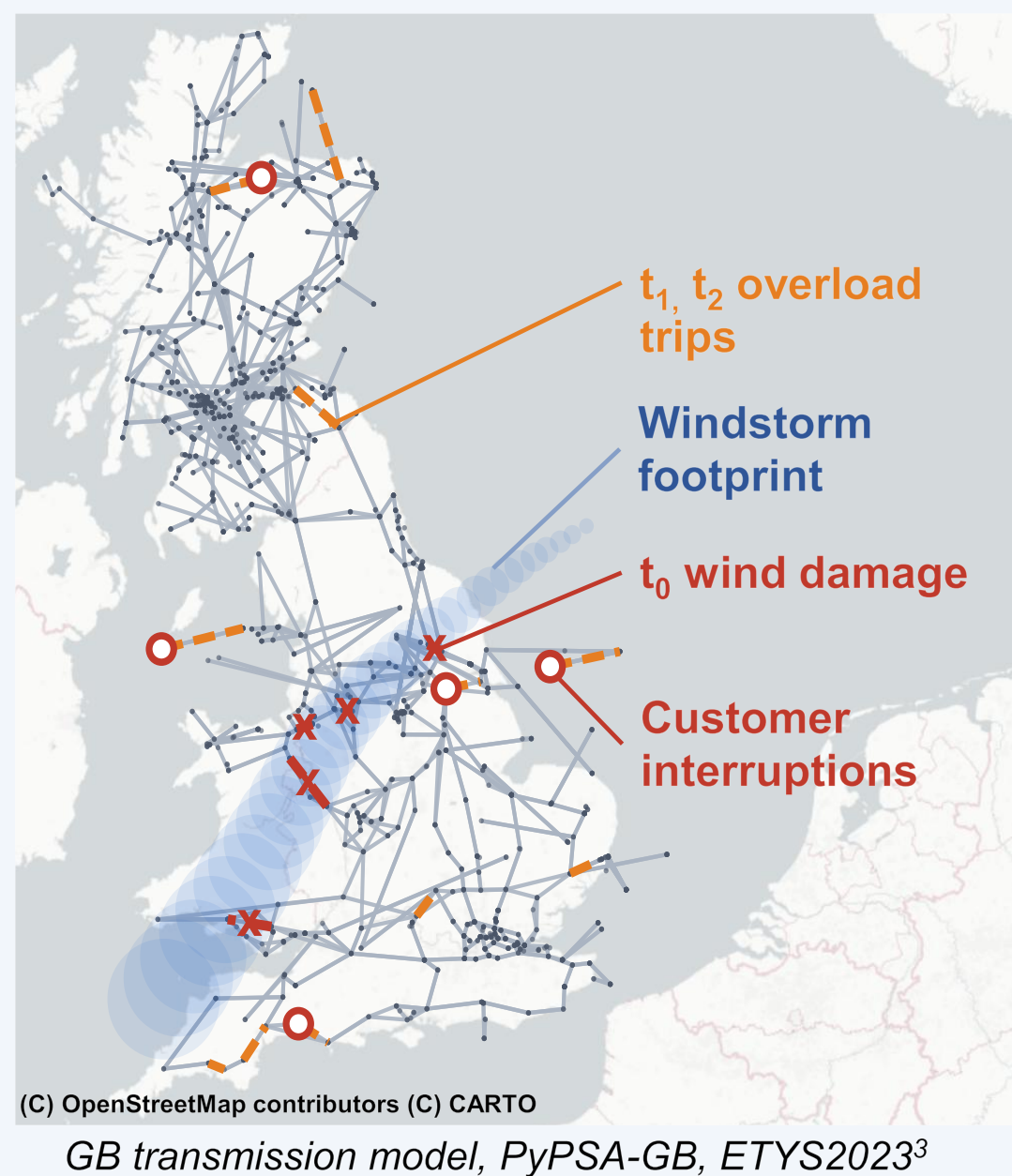


“Most climate projections indicate that winter windstorms will increase slightly in number and intensity over the UK.” — Met Office²

- 📈 **Extremes are rising:** more frequent and more intense storms, floods and other extremes
- ⚡ **Power grid risks:** weather damage can cascade into widespread interruptions.
- 🔧 **Tools needed:** resilience assessment across future climate and network scenarios.

Research Aims

- 1 Simulate**
Windstorm-initiated cascades on a GB transmission network
- 2 Deliver**
Open data, models and workflows on DAFNI
- 3 Reveal**
How investment choices today shape resilience over decades
- 4 Engage**
Operators, regulators and researchers through workshops



Workshop Focus

Future Energy Network Resilience Under Climate Stress

March 2027 | In person, full day | 30 to 40 participants

- Shared scenario matrix**
Climate futures × network evolution pathways
- Live demo and hands-on**
Resilience workflows running on DAFNI
- Structured breakouts**
Data needs, method gaps, investment priorities
- Expert panel**
cross-sector perspectives on operation and planning

References

- EM-DAT, CRED / UCLouvain, Brussels, accessed via Our World in Data, ourworldindata.org
- Met Office, UK and Global Extreme Events: Wind Storms, www.metoffice.gov.uk.
- PyPSA-GB open GB transmission network model, network data from NGESO Electricity Ten Year Statement 2023.

How I Am Using the DAFNI Platform

- 1 Data store**
Host weather hazard and network datasets for open, citable reuse
- 2 Workflow engine**
Run resilience quantification pipeline, with parallel job submission for large simulation tasks.
- 3 Visualisation service**
Publish interactive geospatial dashboards and worked examples of power outage events.

Workflow on DAFNI

Windstorm hazard processing

Gust speeds, return periods and spatial footprints

Component fragility modelling

Weather intensity into component failure probabilities

Cascading failure simulation

Sequential outages propagating through the network

Resilience quantification

Energy not supplied, and resilience trajectories

Geospatial visualisation dashboards

Animated outage maps with scenario comparison

Deliverables Plan

Apr to Jun 2026

Fellowship workplan and stakeholder engagement plan agreed. ✓ **complete**

Jul to Sep 2026

Hazard and network datasets live on DAFNI; presentation at the DAFNI Annual Conference
► **in progress**

Oct to Dec 2026

Simulation workflows, dashboards and tutorial notebooks live on DAFNI

Jan to Feb 2027

Workshop invitations, scenario briefing materials and hands-on exercise built on DAFNI.

Mar 2027

Workshop delivered; summary report and gap analysis published.

Apr to Jun 2027

White paper with technical recommendations; final platform resources.

